

Project Title: Freelancer Marketplace System

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Link to the repository on your GitHub account having EJ Mini project documents and source files:
<https://github.com/vrajghevariya/Freelancer-Marketplace-System>

1. Introduction

The Freelancer Marketplace System is a web application where clients can post projects and freelancers can bid on them. Clients can compare proposals, hire freelancers, track project progress, make secure payments, chat in real time, submit reviews, and download invoices.

Unlike a normal marketplace, this system will also include **AI features** to make it unique.

1. Background:

The freelancing industry has experienced rapid growth due to the increasing demand for remote work and digital services. Businesses and individuals often require skilled professionals for software development, graphic designing, content writing, digital marketing, video editing, and many other services. However, finding reliable freelancers and managing projects efficiently remains a challenge.

The Freelancer Marketplace System is a web-based platform developed using Enterprise Java technologies that connects clients with freelancers in a secure and efficient environment. The system enables clients to post projects, receive proposals, compare freelancers based on skills and ratings, assign work, track project progress, and provide feedback after project completion.

To improve user experience, the system incorporates AI-powered features such as intelligent freelancer recommendations, resume analysis, project description generation, and budget estimation. These capabilities simplify project management, reduce manual effort, and help users make informed decisions. The platform provides transparency, security, and convenience while promoting professional collaboration between clients and freelancers.

1.2 Objectives

The major objectives of the Freelancer Marketplace System are:

- Develop a secure online platform connecting freelancers and clients.
- Enable clients to post projects with detailed requirements.
- Allow freelancers to search available projects and submit proposals.
- Provide secure user authentication and authorization.
- Implement real-time communication between clients and freelancers.
- Track project progress from assignment to completion.
- Maintain project history and payment records.
- Allow users to rate and review completed projects.
- Integrate AI features for freelancer recommendation and project assistance.
- Provide an easy-to-use, responsive, and scalable application using Enterprise Java technologies.

2. Requirements Gathering

Requirement Gathering is the process of identifying and understanding what users expect from the system before development begins. It helps developers understand the features, functions, and constraints of the project. For this project, requirements were collected by studying existing freelancer platforms (such as Fiverr and Upwork), discussing user needs, and analyzing the challenges faced by clients and freelancers.

2.1 Functional Requirements

The system shall provide the following functionalities:

User Module

- User Registration
- Secure Login
- Forgot Password
- Email Verification
- Profile Update

Client Module

- Create Client Profile
- Post New Project
- Edit/Delete Project
- View Bids
- Hire Freelancer
- Track Progress
- Release Payment
- Give Rating

Freelancer Module

- Create Professional Profile
- Upload Resume
- Add Skills
- Browse Projects
- Apply/Bid for Projects
- Upload Completed Work
- View Earnings
- Track Assigned Projects

Project Module

- Project Creation
- Project Categories
- Budget Management
- Deadline Management
- Status Tracking
- File Upload

AI Module

- AI Resume Analysis
- AI Skill Matching
- AI Budget Prediction
- AI Project Description Generator
- AI Chatbot Support

Communication Module

- Chat between Client and Freelancer
- Notifications
- Email Alerts
- Project Updates

Review Module

- Ratings
- Reviews
- Freelancer Ranking

Admin Module

- Manage Users
- Approve Accounts
- Remove Fake Profiles
- View Reports
- Monitor Projects
- Manage Categories

2.2 Non-functional Requirements

Performance

- Support more than 500 concurrent users.
- Dashboard should load within 3 seconds.
- Search results should appear within 2 seconds.

Security

- Password encryption using BCrypt.
- JWT-based authentication.
- Role-Based Access Control (Admin, Client, Freelancer).
- SQL Injection prevention.
- Cross-Site Scripting (XSS) protection.

Reliability

- 99% system availability.

- Automatic database backup.
- Error handling and recovery mechanisms.

Scalability

- Modular architecture using Spring Boot.
- Easy integration of additional AI features and payment gateways.
- Support increasing numbers of users without major performance degradation.

Usability

- User-friendly interface.
- Responsive design for desktop and mobile devices.
- Simple navigation with intuitive dashboards.

Maintainability

- Clean MVC architecture.
- Well-documented source code.
- Easy debugging and future enhancements.

Portability

- Compatible with Windows, Linux, and macOS.
- Runs on modern web browsers such as Chrome, Edge, Firefox, and Safari.

Availability

- 24×7 access through the web.
- Secure cloud deployment support.